
The Condon Report and UFOs

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Scientific Study of Unidentified Flying Objects, a report by Dr. Edward U. Condon, Director of the University of Colorado Project. Bantam Books, New York, in association with the New York Times. 965 pages, including index. \$ 1.95 paper.

Reviewed by

J. Allen Hynek

As consultant on UFOs to the U.S. Air Force for more than 20 years, Dr. Hynek has examined thousands of reports of "flying saucers" and investigated many of them personally. At the beginning of his consultative assignment, his mission was to determine which of the sightings were due to astronomical phenomena - meteors, planets, or stars. By the end of 1949, Dr. Hynek had examined about as many UFO cases as the Condon Report staff has. He came to the same conclusion as Dr. Condon - that the UFO phenomenon was hardly worth serious scientific consideration. In the years since then, however, Dr. Hynek has had reason to change his earlier opinion. He does not agree with the Condon Report and in this review essay he tells why. Dr. Hynek is head of the Department of Astronomy and director of the Lindheimer Astronomical Research Center of Northwestern University.

Physical scientists who know Edward U. Condon through his work in molecular physics and quantum mechanics will find the hand of the master strangely missing in *Scientific Study of Unidentified Flying Objects*. Not only is his talent for organizing and deftly attacking a problem unapparent but, for example, he is not listed as having personally looked into any of the 95 cases to which various members of the rather fluid committee addressed themselves. (Yet his characteristic humor comes through delightfully in his chapter on the recent history of the UFO.)

It is unfortunate that, almost certainly, popular history will henceforth link Dr. Condon's name with UFOs and only the arcane history of physics will accord him his true place and record his brilliant career in contributing to the understanding, with mathematical elegance, of the nature of the physical world. These contributions UFOs cannot take away from him, even though his work with this problem is analagous to that of a Mozart producing an uninspired pot-boiler, unworthy of his talents.

The *Scientific Study of Unidentified Flying Objects* is a strange sort of scientific paper and does not fulfill the promise of its title. Even the color- cover (over which Condon, however, probably had no control) is misleading. Instead of portraying one of the relatively few photographs that remain unidentified we find an immediately identifiable photo of a lens flare.

The report essentially is a collection of case histories and special reports by members of Dr. Condon's staff and investigators working under contract with the University of Colorado. Scientifically trained readers will find these papers as troublesome and dull to read as they probably were to write.

While devoted in the large part to exposing hoaxes or revealing many UFOs as mis-identifications of common occurrences, the book leaves the same strange, inexplicable residue of unknowns which has plagued the U.S. Air Force investigation for 20 years. In fact, the percentage of "unknowns" in the Condon report appears to be even higher than in the Air Force investigation (Project Blue Book) - which led to the

Condon investigation in the first place. Every contributor to the report finds in his particular area of examination (photos, radar-visual sightings, physical evidence, etc.) something that cannot be dismissed as a mis-identification of known phenomena.

One of the contributors, Dr. William K. Hartmann, an astronomer at the University of Arizona, sums up the overall situation as follows: "The present data are compatible with but do not establish either the hypothesis that (1) the entire UFO phenomenon is a product of mis-identification, poor reporting and fabrication, or that (2) a very small part of the UFO phenomenon involves extraordinary events."

An unidentified flying object (UFO, pronounced OOFo) is here defined as the stimulus for a report made by one or more individuals of something seen in the sky (or an object thought to be capable of flight but seen when landed on the earth) which the observer could not identify as having an ordinary natural origin, and which seemed to him sufficiently puzzling that he undertook to make a report of it to police, to government officials, to the press, or perhaps to a representative of a private organization devoted to the study of such objects.

Defined in this way, there is no question as to the existence of UFOs, because UFO reports exist in fairly large numbers, and the stimulus for each report is, by this definition, an UFO. The problem then becomes that of learning to recognize the various kinds of stimuli that give rise to UFO reports.

The UFO is "the stimulus for a report...." This language refrains from saying whether the reported object was a real, physical, material thing, or a visual impression of an ordinary physical thing distorted by atmospheric conditions or by faulty vision so as to be unrecognizable, or whether it was a purely mental delusion existing in the mind of the observer without an accompanying visual stimulus.

Scientific Study of Unidentified Flying Objects, p. 9.

There are other, more provocative statements buried deep within the report. They do not support its overall conclusion that UFO studies do not offer a fruitful field in which to look for major scientific discoveries. Examples are such comments as "unidentified after analysis," or "conceivable but unlikely mis-identification with birds, aircraft, etc."

ELECTRICAL DAMPING PUZZLE

One puzzling aspect of some UFO reports is an electrical damping effect which, according to witnesses, interrupts the ignition and shuts off the engine and lights of a moving automobile. Only one of these cases was examined in the report. The conclusion was "No satisfactory explanation for such effects, if indeed they occurred, is apparent." This reasoning seems to attempt to resolve the problem by dismissing it. One may ask - was it not the function of the investigation to determine whether these reported events did indeed occur? More than 100 cases of electrical or electromagnetic interaction between UFOs and automobiles have been reported, yet the Condon report states: "During the period of field study only one case of automobile engine malfunction came to our attention. There was some ground for skepticism about the report, in that it was made by a diabetic patient who had been drinking and was returning home from a party at 3:00 A.M."

This case is not one of the group I refer to and under the circumstances should have been excluded from the study.

There are other puzzlers described in the report, such as this comment on a UFO sighting claim: "The

residue is a most intriguing report that must certainly be classed as an unknown pending further study, which it certainly deserves. It does appear that this sighting defies explanation by conventional means."

UFOs IN ORBIT

During manned space flights, United States astronauts have reported a number of UFO sightings. One of the Condon group's principal investigators, Franklin Roach, an astronomer, writes: "The three unexplained sightings which have been gleaned from a great mass of reports are a challenge to the analyst."

Over the last 20 years, some of the most baffling cases are those involving radar contacts with as well as visual sightings of the same object. The Condon report does not resolve this long-standing problem. Of one such case, Cordon D. Thayer of the Environmental Science Services Administration, a staff member of Colorado Project, observed: "This must remain as one of the most puzzling radar cases on record and no conclusion is possible at this time. It seems inconceivable that an anomalous propagation (AP) echo would behave in the manner described, even if AP had been likely at the time. In view of meteorological situation it would seem that AP was rather unlikely. Besides, what is the probability that an AP return would appear only once, and at that time appear to execute a perfect ILS (Instrument Landing System) approach?"

Again, a staff report comments: "In conclusion, although conventional or natural explanations certainly cannot be ruled out, the probability of such seems low in this case and the probability that at least one genuine UFO was involved appears to be fairly high."

CHALLENGE TO CURIOSITY

Admittedly, I Have taken these statements out of context and the great bulk of the report over-balances them. But the cases these statements refer to are glaringly there - an outright challenge to human curiosity, the foundation stone of scientific progress. It is difficult to understand why the National Academy of Sciences has fully indorsed Dr. Condon's opinion that no further work on the UFO phenomenon should be done.

As scientific director of the project created to study the vexing problem of UFOs, Dr. Condon undertook a responsibility which may have been distasteful to him from the start. He did so very likely out of a sense of duty, in the same manner that one might, with a deep breath (but through a handkerchief) undertake to sweep out an ill-kept stable. What an Augean stable it was, Condon undoubtedly did not realize, and I feel he grossly underestimated the scope and nature of the problem he was undertaking.

Now, as any scientist would, Dr. Condon defined his terms at the start, but in his very definition of the UFO he fell into a trap. Dr. Condon states, "An unidentified flying object ... is defined as the stimulus for a report made by one or more individuals of something seen in the sky (or an object thought to be capable of flying but seen when landed on the earth) which the *observer* could not identify as having an ordinary natural origin."

The unmanagability of this definition is brought out well in Samuel Rosenberg's chapter, "UFO's in History: "...a report of all such sightings of mysterious objects which the observer 'could not identify' would fill the entire space devoted to the project as a whole." And then some! For, in another section of the report, it is pointed out that perhaps a few as 10 per cent of sightings of UFOs are ever reported. And that percentage relates to this country, while the UFO phenomenon is global. In discussing ancient reports, Rosenberg makes the observation that most everything in the sky was a UFO to pre-scientific man: auroras, lunar halos, rainbows, tornados, lightning - even the sun and the moon. And "what wild guesses were made," continues Rosenberg. Just as today, one might add, guesses are made about things which have not been admitted onto

the playing field of science.

ANATOMY OF A CASE

By adopting so broad a definition of UFO, too much was admitted for the possible study when only limited time and funds were available. let us suppose Condon had adopted this definition instead: A UFO is a report ... the contents of which are puzzling *not only to the observer but to others who have the technical training the observer may lack.*

Why clutter up a study with reports which a cursory examination by people experienced with the subject could almost certainly have dismissed as Venus, a balloon, or a twinkling star? It may be of interest to sociologists that a large percentage of our population cannot identify a bright planet or a bright meteor, but it is of little value to include such trivial cases when others left untouched are truly puzzling (reported effects on car ignition systems, effects on animals and people, cases which have had a traumatic effect on the witnesses, and in some instance, have changed the tenor of their lives, close encounters with craft and blinding lights). Should not the purpose of a study such as Dr. Condon's have been to determine whether there was anything to *truly puzzling* reports - not to obvious cases of trivial mis-identifications?

On the basis of many years experience with the UFO phenomenon, I would have deleted nearly two-thirds of the cases included in the report as potentially profitless for the avowed purposes of the project as stated by Dr. Condon himself: "As indicated by its title, the emphasis of this study has been on attempting to learn from UFO reports anything that could be considered as adding to scientific knowledge." Examining reports that stem from obvious (to anyone with experience in these things) mis-identifications of planets, stars, etc., can add little to scientific knowledge. Far greater care should have been taken in screening cases to be studied, for, as Thurston E Manning, Vice President for Academic Affairs of the University of Colorado, writes: "The reader should thus bear in mind that this study represents the first attempt by a group of highly qualified scientists and specialists to examine coldly and dispassionately ..." What? Misidentifications of Venus, the predicted (by mental telepathy) landing of a UFO, obvious radar chaff, an admitted balloon prank by some students (admitted within hours of receipt of the report by the staff), a smoke ring from a simulated A-bomb explosion at a military installation, the nightly setting of the planets Venus and Saturn, an obvious power outage caused by a short circuit accompanied by bright flashes, a two to three second observation of a flash of light which almost certainly was a meteor? Even a preliminary evaluation of these incidents should have indicated that it was a waste of time to investigate them.

THE "STRANGENESS" INDEX

Over the years I have used a very simple two-dimensional classification method for screening UFO reports for potential scientific value. It is a simple plot of "strangeness" against "credibility of witnesses."

"Strangeness" is a measure of the difficulty of fitting, by scientifically trained persons, the contents of a report to a highly likely physical explanation. Thus, if a bright streak of light is seen to course across the sky in a matter of seconds, there certainly is no reason after even a glance to suggest that the stimulus was anything other than a bright meteor - and one can assign this report a strangeness of 1, or at most, 2.

If, on the other hand, a metallic craft, brilliantly lighted, is reported to have been observed to land, or to have cavorted about the skies in a most "unscientific" manner, this calls for a higher strangeness index. Of course, nothing has been said about believing the contents of the report. But one can, by proper investigation and application of tests, make a meaningful effort to evaluate the witnesses of a UFO "happening" in terms of everyday credibility. Are these "reliable" witnesses; do they pay their debts, are they highly regarded in the community, would they have had any reason to profit from making their report, would it be more likely that they would have suffered by making the report in the first place? Is there anything to indicate that their

emotional nature is such as to cause them to react to perceptual stimuli in a manner to make a "UFO mountain" out of an everyday molehill?

A proper study of the UFO phenomenon for the purposes of assaying potential scientific value implies a preliminary stage in which cases of high strangeness, reported by witnesses of respected standing to their communities, are selected for detailed study. Out of 21 radar-visual cases studied by Thayer, only 3 would I have assigned a, sigma (strangeness) of 4, and none of 5. I would have assigned to the other 18 a sigma of 1, 2, or 3. Admittedly the assignment of these ratings to cases is a matter of individual judgment, but when several independent assessments are made by qualified persons, there is fair agreement, especially as to the "strangeness" of a case; credibility is obviously open to greater variance.

Both the public and the project staff, apparently, have confused the UFO problem with the ETI (extra-terrestrial intelligence) hypothesis. This may hold the greatest popular interest, but it is not the issue. The issue is: Does a legitimate UFO phenomenon exist?

Let us suppose that a committee of nineteenth century scientists had been asked to investigate the phenomenon of the aurora as a single project. It would not have been responsible to state that the polar phenomenon gave no evidence of the existence of some meta-terrestrial intelligence. The issue would have been whether the aurora could be explained in terms of nineteenth century physics.

It may be that UFO Phenomena are just as inexplicable in terms of twentieth century physics. From this point of view, how does the Condon Report serve science when it suggests that a phenomenon which has been reported by many thousands of people over so long a time is unworthy of further scientific attention?

Investigative experience over the last 20 years has indicated to me that the UFO phenomenon, if it be physically real, is a *rara avis*. I suggest that of all the cases studied in the report, the following might be truly worthy of study in depth: Cases 2, p. 248; 5, p. 260; 10, p. 277; 46, p. 396; 57, p. 469; 19-B, p. 161; 14 Na, p. 127; 14-Nb, p. 128; unnumbered case, p. 139; 1482-N, p. 143 and unnumbered case, p. 236.

While it was perhaps laudable to ask an untried, and therefore, presumably, unbiased group to take a fresh look at the UFO problem, this procedure was akin to asking a group of culinary novices to take a fresh look at cooking and then open a restaurant. Without seasoned advice, there would be many burned pots, many burned fingers, many dissatisfied customers.

Much of the time of the project personnel, it appears to me, was spent in groping for a methodology. It appears also that graduate students did the yeoman part of the investigations in the relatively few field trips made, a result, undoubtedly, of limited funds.

Finally, in the matter of methodology a philosopher of science would find a serious operational and epistemological flaw: An hypothesis which covers everything covers nothing. Let us state this in the form of a UFO theorem: For any given reported UFO case, if taken by itself and without respect and regard to correlations with other truly puzzling reports in this and other countries, a possible natural, even though far-fetched, explanation can always be adduced. This is so if one operates solely on the hypothesis that all UFO reports, by the very nature of things as we know them, must result from well known and accepted causes.

It follows as a corollary that it would have been impossible for the Condon investigation to have regarded any report as arising from anything other than natural causes, a hoax, or a hallucination. Thus, for instance, we have this astonishing analysis (Case not numbered, p. 140): "This unusual sighting should therefore be assigned to the category of some almost certainly natural phenomenon which is so rare that it apparently has

never been reported before or since."

Obviously this statement could be made of any puzzling case. Or, (p. 164, Case 2): "In summary, this is the most puzzling and unusual case in the radar-visual file. The apparently rational, intelligent behavior of the UFO suggests a mechanical device of unknown origin as the most probable explanation of the sighting. However, in view of the inevitable fallibility of witnesses, more conventional explanations of this report cannot be entirely ruled out." in Case 46 (p. 407) the investigator is hard pressed, but still applies the theorem: "This is one of the few UFO reports in which all factors investigated, geometric, psychological, and physical appear to be consistent with the assertion that an extraordinary flying object, silvery, metallic, disk-shaped, tens of meters in diameter, and evidently artificial, flew within sight of two witnesses. It cannot be said that the evidence positively rules out a fabrication, although there are some physical factors such as the accuracy of certain photometric measures of the original negatives which argue against a fabrication." Final verdict: "fabrication."

Final judgment of the work of the Condon Committee, which was not a study of truly Unidentified Flying Objects, but largely of easily identifiable objects will be handed down by the UFO phenomenon itself. Past experience suggests that it cannot be readily waved away.

There is, however, one area in which the reviewer is in accord with Dr. Condon, and that is in his recommendation that science credit not be given in elementary schools for term papers and projects on UFOs. School children are too lacking in critical faculties to be turned loose in UFO land. Present material available to them is apt to be pulp "literature," itself written sensationally and uncritically, cases undocumented, with no attention whatever to analysis; a mere collection of sensationalized anecdotes.

If the Condon Report helps to clear away the miasma of pseudo-science, wishful thinking, and sensationalism in this area, the stage may yet be prepared for a more effective study of the strange and perplexing phenomenon of UFOs. To this end, care should be taken that the files of the Condon Committee not be destroyed, as reportedly were the data in a 1953 investigation of UFOs by another Air Force contractor whose identity was classified and whose data led to Report No. 14 of Project Blue Book.



[Return to home](#)



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